5. Rational Function Fields

Jahresbericht der DMV 22 (1913), pp. 316–319

The following questions originally go back to conversations with E. Fischer. Some of the questions, moreover, were already raised and settled by E. Steinitz for the special case of one indeterminate — and indeed under more general assumptions on the coefficient field.[1]

By a "rational function field" I understand a field whose elements are rational functions of n indeterminates, with coefficients from a prescribed number field, which in particular may also comprise all complex numbers. Examples of such fields are the field of symmetric functions of n quantities, or more generally the totality of rational functions of n indeterminates that allow the permutations of a certain group (Lagrange's *Gattungsbereiche*); furthermore, the invariant field.

Between the first two fields just mentioned and the latter there is a characteristic difference: the former contain n algebraically independent functions, the elementary symmetric functions; whereas the number of algebraically independent invariants is always smaller than the number of indeterminates. It is therefore important that, in general, one can associate such fields of the second kind one-to-one with fields of the first kind, by replacing some of the indeterminates by numbers. In what follows I shall therefore restrict myself to fields of the first kind, i.e. to fields with n algebraically independent functions; by virtue of the association, the results then hold in general.

The questions group themselves around three basis concepts: rational basis, minimal basis, and integrality basis.

1. By a "rational basis" I understand a finite number of functions of the field such that every function of the field can be represented as a rational combination of this finite number, with coefficients from the prescribed coefficient field. By simple considerations — which in the case of one indeterminate are also found in E. Steinitz, loc. cit. — one proves the existence of a rational basis for every rational function field. For example, by Lagrange's theorem, the elementary symmetric functions and one function belonging to the group form a rational basis for the Lagrange *Gattungsbereiche* mentioned earlier.

2. Every rational basis must contain (for fields of the first kind) at least n functions; if it contains exactly n functions, which then must be algebraically independent, I call it a "minimal basis." The question of the existence of a minimal basis can be answered in part by theorems of Lüroth, Castelnuovo, and Enriques.[2] According to these, for fields of one indeterminate there always exists a minimal basis: the Lüroth function of the field, uniquely determined up to fractional linear transformation (cf. Steinitz § 24). For fields of two indeterminates, a minimal basis exists always, and in general only, if one permits an algebraic extension of the coefficient field, whereas for three and more indeterminates a minimal basis in general does not exist at all. No attempt has yet been made to characterize the special fields with minimal basis.

I have pursued further the question of the minimal basis for Lagrange's *Gattungsbereiche*. Here it acquires the following meaning: "If the Lagrange *Gattungsbereich* belonging to the group G possesses a minimal basis, then one can construct rationally, by a parametric representation, the totality of equations with prescribed group G; and indeed for every number field that contains the coefficient field of the minimal basis — hence in particular for any arbitrary number field if this coefficient field is the field of rational numbers."

The simplest example is provided by the symmetric group; here the elementary symmetric functions form a minimal basis with rational numerical coefficients. From this one obtains as parametric representation simply the equation with indeterminate coefficients. Hilbert's irreducibility theorem shows how one passes from this to arbitrarily many affectless equations over every number field.

Now the existence of the minimal basis for all groups occurring in equations of degrees 3 and 4 follows directly from the theorems of Lüroth and Castelnuovo; at the same time it further appears that this minimal basis has rational numerical coefficients. Thus, for every arbitrary number field, one can rationally construct the totality of equations of degrees 3 and 4 with prescribed group. For the cyclic and dihedral groups there exists a minimal basis with the field of the nth roots of unity as coefficient field; the same holds for a few further metacyclic groups. The question whether the minimal basis is at all characteristic for certain groups must remain undecided.

[1] E. Steinitz: Algebraische Theorie der Körper, especially § 24. Crelle's Journal 137. 1910.

[2] J. Lüroth: Beweis eines Satzes über rationale Kurven. Math. Ann. 9. 1875. — G. Castelnuovo: Sulla razionalità delle involuzioni piane. Math. Ann. 44. 1893. — F. Enriques: Sopra una involuzione non razionale dello spazio. Rend. Acc. Linc. vol. 21. 21 Jan. 1912.

3. To arrive at the last basis concept, I consider the totality of polynomials contained in the field. By an "integrality basis" I understand a finite number of these polynomials such that every polynomial of the field can be represented as an integral rational combination of this finite number, with coefficients from the prescribed coefficient field. The question of the existence of an integrality basis — which, for example, contains as a special case the finiteness of the invariant system — was posed by Hilbert in his "Mathematical Problems"[3] in a somewhat different formulation, as the problem of relatively integral functions.

For the moment I can indicate only one class of fields for which an integrality basis exists. Namely, if the field contains a system of n polynomials such that the homogeneous resultant of the terms of highest dimension of these polynomials is different from zero, then an integrality basis exists; it is given by the coefficients of all u in the equation, irreducible in the field, for the linear form:

$$u_0 = u_1 x_1 + u_2 x_2 + \cdots + u_n x_n.$$[4]

If, in particular, the value of the resultant is equal to a unit of the coefficient field, then the integrality of the representation is also secured. An example of this class of fields is furnished by Lagrange's *Gattungsbereiche*, since the resultant of the elementary symmetric functions has the value ± 1. A further example (without integrality) is given by all fields that contain only one algebraically independent polynomial; here the integrality basis is identical with the Lüroth function of the smallest subfield containing all polynomials. Thus, as is well known, all integral rational projective invariants of a quadratic form in n variables are integral functions of the discriminant.

The condition just given is only sufficient, by no means necessary, for the existence of an integrality basis. One can easily construct fields in which the resultant of every n polynomials vanishes, and which nevertheless possess an integrality basis given by the coefficients of that irreducible equation. The question arises whether perhaps even in the most general case the coefficients of that equation furnish the integrality basis.

[3] D. Hilbert: Mathematische Probleme. Lecture, Paris 1900. Problem 14. Göttinger Nachr. 1900.

[4] This irreducible equation is found, in the case of one indeterminate, in E. Steinitz's proof of Lüroth's theorem.

6. Fields and Systems of Rational Functions

Math. Ann. 76 (1915), pp. 161–196

The present paper treats basis questions for arbitrary systems of rational and integral rational functions. The methods of field theory used here make the settlement of these questions for fields of rational functions — rational function fields[5] — appear as the essential matter, while the generalization of the results to arbitrary systems arises as a consequence.

Among basis questions for general systems, up to now only the existence of a module basis, guaranteed by Hilbert's theorem (Math. Ann. 36), has been known. In what follows, the question of rational representability is answered completely by the existence of a rational basis for every arbitrary system (§ 7); the rational basis of fields is already obtained in § 4. This existence of the rational basis permits us throughout to start from the abstractly defined field or system, and thereby to avoid difficulties that are caused only by the special choice of the rational basis and not by the system itself, such as the occurrence of special denominators or of fundamental points of the basis functions.

For fields the question of the minimal basis is also treated, i.e. of a rational basis consisting of algebraically independent functions (§ 6). The methods of field theory further lead to a distinguished rational basis, the involution basis[6] (§ 5), which becomes especially important for integrality domains consisting of polynomials (§ 8). In that setting it gives a representation with a fixed denominator (a power of a function determined by the integrality domain), as is known for example in the special case of the typical representation of invariants.

From rational representability we now draw, as far as possible, consequences for integral rational representability, i.e. for the question of finiteness in the narrower sense.

The finiteness theorems known up to now all rest on the fact that Hilbert's theorem on module bases guarantees finiteness for all systems in which a representation

$$F = A_1 f_1 + A_2 f_2 + \cdots + A_k f_k$$

entails a second representation

$$F = B_1 f_1 + B_2 f_2 + \cdots + B_k f_k,$$

where the B_i belong to the system and are of lower degree than F.

By contrast, the theorem on the rational basis and the methods of field theory permit one to infer finiteness under assumptions of an entirely different kind.

Thus, as a partial answer to a problem of Hilbert (Mathematical Problems 14), one obtains a class of relatively integral functions (relatively integral domains of the first kind) which are finite integrality domains and whose integrality basis is given by the involution basis mentioned above (§ 10). In analogy with Hilbert's proof of Kronecker's theorem on the fundamental system of algebraic integers (Complete invariant systems, § 2; Math. Ann. 42), it can further be shown that all regular systems of polynomials — i.e. all systems that can be mapped to ones without fundamental points — possess an integrality basis (§ 12), whereas nonregular systems without integrality basis can easily be specified. As a simple example of this fact one may mention the theorem: "In every arbitrary system of infinitely many polynomials in one indeterminate, there exists a finite number of these polynomials such that every polynomial of the system can be represented as an integral rational combination of this finite number." Further examples are found in § 13.

Finally, the last paragraphs (§§ 14 and 15) give a sharpening of the basis theorems with respect to integrality.

An essential tool of the investigation is the transfer principle of § 3, according to which it suffices to consider all basis questions raised here for those systems in which the number of indeterminates coincides with the number of algebraically independent functions of the system.

The impetus for the present work came from conversations with Mr. E. Fischer, in particular from a question of his concerning the minimal basis of Lagrange's *Gattungsbereiche*. The investigations relating to this, which led to the construction of equations with prescribed group (cf. the cited communication on "Rational Function Fields," part 2), are reserved for a further publication.

[5] Cf. a preliminary communication on the occasion of the Vienna meeting of natural scientists: Rationale Funktionenkörper, Jahresber. d. D. Math.-Ver. 22 (1913), where an overview is given of the questions and of the results concerning function fields.

[6] The name was chosen because, in its geometric interpretation, the rational function field represents the most general involution in a linear space.

The essential extension of the present paper over the original communication consists in the fact that the basis theorems stated there only for fields or special integrality domains are here transferred to arbitrary systems. This transfer succeeds in a simple way by means of the concept of the smallest containing field of an arbitrary system, as a generalization of the quotient field of an integrality domain (§ 7). I was led to form this concept by the fact that Mr. K. Hentzelt, after learning the rational basis of fields, was able to prove the existence of the rational basis for arbitrary systems by a detour through Hilbert's module basis. I was also led to the theorem on the integrality basis of regular systems by K. Hentzelt's observation that the arguments I had applied to integrality domains were valid for arbitrary systems subject to the same assumptions; likewise I owe to Mr. Hentzelt a few isolated smaller remarks.

Finally, let it be mentioned that in the case of one indeterminate the rational basis and minimal basis of fields are found in E. Steinitz's "Algebraische Theorie der Körper," § 24 (Crelle's Journal 137); there the coefficient field is taken to be a field in the most general sense. The question how far the theorems given here remain valid under these more general assumptions is not touched on in what follows; in any event the proofs require modification, since, for example, the tools from the theory of functional determinants fail for fields of characteristic p.

§ 1. Fields $\mathfrak{K}_{n\rho}$ and systems $\mathfrak{S}_{n\rho}$.

In what follows we are concerned with the investigation of systems of rational functions in n indeterminates, especially fields of rational functions.

By $f(x_1,\ldots,x_n)$, $g(x_1,\ldots,x_n)$, …, or abbreviated $f(x)$, $g(x)$, …, one should therefore always understand rational functions of $x_1,\ldots,x_n$; these are assumed to be in reduced representation — that is, numerator and denominator relatively prime. Integral rational functions (polynomials) will be denoted by capital letters, $F(x)$, $G(x)$, …. The coefficient field Ω is assumed to be some number field; in particular, it may also contain all complex numbers. The field Ω may also contain a finite number of parameters.

The quantities from Ω, hence all functions of degree zero, are always counted as belonging to the system.

With these stipulations, the field consisting of rational functions — the rational function field — can be defined abstractly.

Definition I: A system of rational functions is called a field if it satisfies the conditions:

1. *Together with $f(x)$, and for every quantity c from Ω, it also contains $c\cdot f(x)$.*
2. *Together with $f(x)$ and $g(x)$, it always also contains $f(x)+g(x)$, $f(x)\cdot g(x)$ and — for $g(x)\neq 0$ — the quotient $f(x):g(x)$.*[7]

Special fields are those which arise by adjoining a finite number of rational functions

$$f_1(x),\ f_2(x),\ldots,f_k(x)$$

to Ω; they will be denoted by

$$\Omega(f_1(x),\ldots,f_k(x)) \quad \text{or} \quad \Omega(f_1,\ldots,f_k).$$

[8] Among these special fields we also mention the field arising by adjoining $x_1,\ldots,x_n$ to Ω,

$$\Omega(x_1,\ldots,x_n),$$

which is identical with the totality of rational functions of $x_1,\ldots,x_n$ with coefficients from Ω.

We also introduce (following E. Steinitz) the concept of an intermediate field:

> "A field $\mathfrak{M}$ lies between Ω_1 and Ω_2 if it contains Ω_1 and is contained in Ω_2;"

then Definition I also admits the following formulation:

The field of rational functions of n indeterminates is an intermediate field between Ω and $\Omega(x_1,\ldots,x_n)$ which effectively contains the n indeterminates in at least one function.

Beside the number of indeterminates, there arises for systems of rational functions a further characteristic number, the number of algebraically independent functions, or the algebraic rank, by the following definition.

[7] Here $f(x)$ and $g(x)$ may also be of degree zero, i.e. quantities from Ω.

[8] That these "special fields" already exhaust the totality of all fields follows in § 4.

Definition II: A system of rational functions has algebraic rank ρ if ρ functions of the system can be specified that are algebraically independent, but every $(\rho+1)$ functions of the system are algebraically dependent.[9]

Systems of (finitely or infinitely many) rational functions in n indeterminates and of algebraic rank ρ will be denoted by the double index

$$\mathfrak{S}_{n\rho}.$$

In particular, by

$$\mathfrak{K}_{n\rho}$$

one is to understand a field of n indeterminates and of algebraic rank ρ. The inequality holds:

$$1 \leq \rho \leq n.$$

We give a few more examples of fields $\mathfrak{K}_{nn}$ and $\mathfrak{K}_{n\rho}$:

1. Fields $\mathfrak{K}_{nn}$ are Lagrange's *Gattungsbereiche*, which can be defined as

> "the totality of rational (and integral rational) functions of $x_1, \dots, x_n$ that allow the permutations of a permutation group G of $x_1, \dots, x_n$."

They always contain the field of symmetric functions, hence also the n algebraically independent elementary symmetric functions.

2. Fields $\mathfrak{K}_{n\rho}$ are the projective invariant fields, which are formed from the totality of the rational (and integral rational) invariants of one or more ground forms.[10] Here $\rho < n$, since the number of algebraically independent invariants is always smaller than the number of coefficients.

The corresponding statement holds for the invariant fields with respect to subgroups of the projective group.

§ 2. Auxiliary theorem on algebraically dependent functions.

The auxiliary theorem of this paragraph will show in § 3 that the algebraic rank of a system is its properly characteristic number. Namely, the auxiliary theorem shows that by a successive numerical specialization of some indeterminates, such that ρ algebraically independent functions remain algebraically independent, an arbitrary function algebraically dependent on these ρ functions can neither vanish identically nor become indeterminate.

Let therefore the system

$$g_1(x),\ g_2(x), \dots, g_\rho(x) \tag{1}$$

have algebraic rank ρ, where $\rho < n$, and let $h(x)$ be a rational function algebraically dependent on the functions (1). By known theorems, the rank of the functional matrix of (1) is equal to ρ. The indeterminates may therefore be named, for instance, so that

$$\frac{\partial(g_1, g_2, \dots, g_\rho)}{\partial(x_1, x_2, \dots, x_\rho)} = \frac{\Gamma(x)}{G(x)^{\rho+1}} \neq 0, \tag{2}$$

where $G(x)$, as usual, denotes the common denominator of the functions (1).

Now choose the number a so that the product $G(x) \cdot \Gamma(x)$ is not divisible by $(x_i - a)$, where i is understood to be one of the numbers $\rho+1, \rho+2, \dots, n$. Thus, for $x_i = a$, the common denominator of the functions (1) cannot vanish, and the ρ functions

$$[g_1(x)]_{x_i=a}, \dots, [g_\rho(x)]_{x_i=a} \tag{3}$$

are likewise algebraically independent. As we shall say, the algebraic rank of the system (1) is preserved under the substitution $x_i = a$.

[9] ρ functions $f_1(x), \dots, f_\rho(x)$ are called algebraically independent if from every relation

$$F\big(f_1(x), \dots, f_\rho(x)\big) = 0 \quad \text{identically in } x_1, \dots, x_n$$

it follows that

$$F(\lambda_1, \dots, \lambda_\rho) = 0 \quad \text{identically in } \lambda_1, \dots, \lambda_\rho.$$

In the opposite case they are called algebraically dependent.

[10] It is expedient to consider only transformations of determinant 1; then every rational combination of arbitrarily many invariants again admits the transformation, and one does indeed obtain a field. The homogeneous, isobaric invariants taken by themselves do not form a field, but they do form a system $\mathfrak{S}_{n\rho}$.

Let the irreducible equation satisfied by $h(x)$ with respect to the functions (1) be given by

$$\begin{aligned} A_0(g_1,\ldots,g_\rho)\,h(x)^\tau + A_1(g_1,\ldots,g_\rho)\,h(x)^{\tau-1} + \cdots \\ + A_\tau(g_1,\ldots,g_\rho) = 0, \end{aligned} \tag{4}$$

where $A_0(g_1,\ldots,g_\rho)$ and $A_\tau(g_1,\ldots,g_\rho)$ do not vanish identically. In order to pass from this to an identity between polynomials, we set

$$g_i(x) = G_i(x) : G(x); \qquad h(x) = H_1(x) : H_2(x), \tag{5}$$

and note that — because of the assumed reduced representation of $h(x)$ — $H_1(x)$ and $H_2(x)$ are relatively prime. By virtue of (5), (4) becomes

$$\begin{aligned} &B_0(G,G_1,\ldots,G_\rho)\,G(x)^{\sigma_0}H_1(x)^\tau \\ &\quad + B_1(G,G_1,\ldots,G_\rho)\,G(x)^{\sigma_1}H_1(x)^{\tau-1}H_2(x) + \cdots \\ &\quad + B_\tau(G,G_1,\ldots,G_\rho)\,G(x)^{\sigma_\tau}H_2(x)^\tau = 0, \end{aligned} \tag{6}$$

where the B_k are homogeneous forms in G, G_i, and at least one of the nonnegative exponents σ_k must be zero.

Suppose now that $H_1(x)$ were divisible by $(x_i - a)$. Then, by (6),

$$B_\tau(G,G_1,\ldots,G_\rho)\,G(x)^{\sigma_\tau}H_2(x)^\tau$$

would also be divisible by $(x_i - a)$. This is excluded by assumption for $G(x)$; hence from the divisibility of B_τ there would also follow the divisibility of

$$A_\tau = B_\tau : G^\lambda,$$

and between the functions (3) there would exist the relation $A_\tau = 0$, contrary to the assumed algebraic independence. Finally $H_2(x)$, being relatively prime to $H_1(x)$, also cannot be divisible by $(x_i - a)$;[11] thus the assumption that $H_1(x)$ is divisible by $(x_i - a)$ leads to a contradiction. In the same way one shows that $H_2(x)$ also cannot be divisible by $(x_i - a)$. Hence the function $[h(x)]_{x_i=a} = h'(x)$ can neither vanish identically nor become indeterminate.

The above conclusion can now be applied again to the system (3) and the function $h'(x)$, again assumed to be in reduced representation, and continuation of the procedure leads finally to the following auxiliary theorem.

Auxiliary theorem: The two systems

$$g_1(x),\ g_2(x),\ldots,g_\rho(x),$$

$$g_1(x),\ g_2(x),\ldots,g_\rho(x),\ h(x)$$

shall each have algebraic rank ρ, where $\rho < n$, and suppose, for instance, that

$$\frac{\partial(g_1,\ldots,g_\rho)}{\partial(x_1,\ldots,x_\rho)} = \frac{\Gamma(x)}{G(x)^{\rho+1}} \neq 0.$$

The numbers

$$a_n,\ a_{n-1},\ldots,a_{\rho+1}$$

are to be chosen so that, under the substitution

$$x_n = a_n, \quad x_{n-1} = a_{n-1},\ldots, \quad x_{\rho+1} = a_{\rho+1},$$

[11]This last conclusion, resting on the reduced representation, would no longer be possible for a substitution $x_i = a$, $x_k = b$ while the remaining assumptions are preserved. Then $H_1(x)$ and $H_2(x)$ could vanish simultaneously, and the quotient would become indeterminate under the substitution, equal to $\frac{0}{0}$; for example

$$h(x) = \frac{x_3(\alpha x_1 + \beta x_2) + x_4(\gamma x_1 + \delta x_2)}{x_3(\alpha' x_1 + \beta' x_2) + x_4(\gamma' x_1 + \delta' x_2)}$$

under the substitution $x_3 = 0$, $x_4 = 0$.

the product $G(x) \cdot \Gamma(x)$ does not vanish — that is, the algebraic rank of the system (1) is preserved. In the successive substitutions

$$[h(x)]_{x_n=a_n} = h^{(n-1)}(x), \qquad [h^{(n-1)}(x)]_{x_{n-1}=a_{n-1}} = h^{(n-2)}(x), \ldots,$$
$$[h^{(\rho+1)}(x)]_{x_{\rho+1}=a_{\rho+1}} = h^*(x_1, x_2, \ldots, x_\rho),$$

let the functions $h(x)$, $h^{(n-1)}(x)$, $\ldots$, $h^{(\rho+1)}(x)$ each be put into reduced representation. Under these assumptions, in the function $h^*(x_1, \ldots, x_\rho)$ neither numerator nor denominator can vanish identically, and the function also cannot become $\frac{0}{0}$.[12]

§ 3. One-to-one mapping of systems $\mathfrak{S}_{n\rho}$ onto systems $\mathfrak{S}^*_{\rho\rho}$.

From the auxiliary theorem we shall immediately obtain a mapping of the systems $\mathfrak{S}_{n\rho}$ onto systems $\mathfrak{S}^*_{\rho\rho}$, by replacing some of the indeterminates by numbers; the algebraic rank ρ thereby presents itself as the properly characteristic number.

Since $\mathfrak{S}_{n\rho}$ has algebraic rank ρ, there exist ρ functions from $\mathfrak{S}_{n\rho}$,

$$g_1(x), \ldots, g_\rho(x), \tag{1}$$

which are algebraically independent. Further, if $f(x)$ denotes an arbitrary function from $\mathfrak{S}_{n\rho}$, then the system

$$g_1(x), \ldots, g_\rho(x),\ f(x) \tag{2}$$

also has algebraic rank ρ, and the systems (1) and (2) satisfy the conditions of the auxiliary theorem.[13]

If, therefore, one determines the numbers

$$a_n,\ a_{n-1}, \ldots, a_{\rho+1}$$

according to the auxiliary theorem in such a way that the algebraic rank ρ of the system (1) is preserved by the substitution — which is possible in arbitrarily many ways — then in the function defined by

$$[f(x)]_{x_n=a_n} = f^{(n-1)}(x), \qquad [f^{(n-1)}(x)]_{x_{n-1}=a_{n-1}} = f^{(n-2)}(x); \ldots,$$
$$[f^{(\rho+1)}(x)]_{x_{\rho+1}=a_{\rho+1}} = f^*(x_1, \ldots, x_\rho),$$

namely

$$f^*(x_1, \ldots, x_\rho), \tag{3}$$

neither numerator nor denominator can vanish identically.

Now let $f(x)$ run through all (finitely or infinitely many) functions from $\mathfrak{S}_{n\rho}$. We consider the system consisting of the totality of all functions (3); it has the following properties:

1. It has algebraic rank ρ; for it contains the functions arising from (1),

$$g_1^*(x_1, \ldots, x_\rho), \ldots, g_\rho^*(x_1, \ldots, x_\rho),$$

which, because of the choice of $a_n, a_{n-1}, \ldots, a_{\rho+1}$, are algebraically independent. The system is therefore to be denoted by

$$\mathfrak{S}^*_{\rho\rho}.$$

14

2. The correspondence of the functions $f(x)$ and $f^*(x)$ is without exception one-to-one.

For if $f_1(x)$ and $f_2(x)$ belong to $\mathfrak{S}_{n\rho}$, then their difference

$$d(x) = f_1(x) - f_2(x)$$

[12] For example, in the function $h(x)$ of the preceding note, the successive substitution gives

$$[h(x)]_{x_4=0} = h^{(3)}(x) = \frac{\alpha x_1 + \beta x_2}{\alpha' x_1 + \beta' x_2}, \qquad h^*(x_1, x_2) = \frac{\alpha x_1 + \beta x_2}{\alpha' x_1 + \beta' x_2}.$$

[13] If necessary, the indeterminates are to be renumbered.

[14] Correspondingly, by $f^*(x)$, $g^*(x)$, $\ldots$ one should throughout understand mapping functions, i.e. functions of $x_1, \ldots, x_\rho$.

is also algebraically dependent on the system (1); moreover

$$d^*(x) = f_1^*(x) - f_2^*(x).$$

Since $d(x)$, together with the system (1), satisfies the conditions of the auxiliary theorem, it follows from

$$d(x) \neq 0$$

necessarily that

$$d^*(x) \neq 0;$$

in other words, different functions from $\mathfrak{S}_{n\rho}$ correspond to different functions from $\mathfrak{S}_{\rho\rho}^*$.

3. From every relation between functions from $\mathfrak{S}_{\rho\rho}^*$,

$$F(f_1^*(x), f_2^*(x), \ldots, f_k^*(x)) = 0 \quad \text{identically in } x_1, \ldots, x_\rho,$$

it follows, for the uniquely corresponding functions from $\mathfrak{S}_{n\rho}$, that

$$F(f_1(x), f_2(x), \ldots, f_k(x)) = 0 \quad \text{identically in } x_1, \ldots, x_n.$$

For with $f_1(x), f_2(x), \ldots, f_k(x)$, every rational combination of these functions is also algebraically dependent on the system (1).

It further follows from

$$\begin{aligned} h(x) &= F(f_1(x), \ldots, f_k(x)) : \\ h^*(x) &= F(f_1^*(x), \ldots, f_k^*(x)). \end{aligned} \tag{4}$$

By the auxiliary theorem, therefore,

$$h(x) \neq 0 \quad \text{implies} \quad h^*(x) \neq 0;$$

q.e.d.

In summary, we obtain

Theorem I: To every system $\mathfrak{S}_{n\rho}$ of (finitely or infinitely many) rational functions there can be assigned — in arbitrarily many ways — a one-to-one system $\mathfrak{S}_{\rho\rho}^*$, in such a way that all relations existing between functions from $\mathfrak{S}_{\rho\rho}^*$ are also preserved in $\mathfrak{S}_{n\rho}$ and conversely. In particular, by (4), a field $\mathfrak{K}_{n\rho}$ again corresponds to a field $\mathfrak{K}_{\rho\rho}^*$.

From Theorem I we draw the consequence that simplifies all further proofs: *properties of systems $\mathfrak{S}_{n\rho}$ which are characterized by finitely or infinitely many relations between functions of the system hold for the systems $\mathfrak{S}_{n\rho}$ as soon as they hold for a mapping system $\mathfrak{S}_{\rho\rho}^*$.*

We call this consequence briefly the "transfer principle."

The simplest example of this mapping is furnished by the theory of algebraic invariants. In fact, by a linear transformation one can always numerically fix some coefficients of the ground form in such a way that all relations valid for the special form also remain valid in general.

§ 4. Rational basis of the fields $\mathfrak{K}_{n\rho}$.

In the following investigations, grouped around basis concepts, we shall consistently use the "transfer principle" of § 3. We first prove the existence of the basis for fields, but state the definitions generally:

Definition III: A finite number of functions from $\mathfrak{S}_{n\rho}$ is called a rational basis if every function from $\mathfrak{S}_{n\rho}$ can be represented as a rational combination of this finite number, with coefficients from Ω.

The rational basis must contain ρ algebraically independent functions, since the algebraic rank of a system derived from a rational basis is equal to the algebraic rank of the rational basis.

We first show that the existence proof for the rational basis admits the transfer principle. Namely, Definition III can be formulated as follows:

The functions from $\mathfrak{S}_{n\rho}$

$$g_1(x),\ g_2(x), \ldots, g_k(x)$$

form a rational basis if and only if the infinitely many relations

$$f(x) = \gamma(g_1(x), \ldots, g_k(x)) \tag{1}$$

are satisfied, where $f(x)$ runs through all functions from $\mathfrak{S}_{n\rho}$ and γ denotes a rational function of k arguments, varying with f.

Thus, if in the mapping system $\mathfrak{S}^*_{\rho\rho}$ a rational basis is given by

$$g^*_1(x),\ g^*_2(x), \ldots, g^*_k(x),$$

which entails the relations

$$f^*(x) = \gamma(g^*_1(x), \ldots, g^*_k(x)),$$

then by Theorem I the relations (1) are fulfilled for $\mathfrak{S}_{n\rho}$. Thus the *transfer principle for the rational basis* says:

From the existence of a rational basis for a mapping system $\mathfrak{S}^*_{\rho\rho}$ follows the rational basis for the original system $\mathfrak{S}_{n\rho}$.[15]

To prove the existence of the rational basis of all fields $\mathfrak{K}_{n\rho}$, we may therefore restrict ourselves to fields $\mathfrak{K}^*_{\rho\rho}$; here a direct generalization of an argument used by E. Steinitz leads to the goal.[16]

Let

$$g^*_1(x_1, \ldots, x_\rho), \ldots, g^*_\rho(x_1, \ldots, x_\rho) \tag{2}$$

be a system of ρ algebraically independent functions from $\mathfrak{K}^*_{\rho\rho}$. By eliminating $x_1, \ldots, x_\rho$ one obtains ρ relations:

$$\begin{aligned} &F_1(g^*_1(x), \ldots, g^*_\rho(x), x_1) = 0, \ldots, \\ &F_\rho(g^*_1(x), \ldots, g^*_\rho(x), x_\rho) = 0, \end{aligned} \tag{3}$$

where in each relation $F_i = 0$ the indeterminate x_i actually occurs; otherwise, contrary to the assumption, there would be an algebraic dependence among the functions (2). The relations (3) say that

$$x_1, x_2, \ldots, x_\rho$$

are algebraic elements with respect to

$$\Omega(g^*_1(x), \ldots, g^*_\rho(x)).$$

Hence the field arising by adjoining $x_1, \ldots, x_\rho$,

$$\Omega(g^*_1, \ldots, g^*_\rho, x_1, \ldots, x_\rho) = \Omega(x_1, \ldots, x_\rho),$$

is a finite algebraic extension of $\Omega(g^*_1, \ldots, g^*_\rho)$. By a known theorem,[17] $\mathfrak{K}^*_{\rho\rho}$, as an intermediate field between $\Omega(g^*_1, \ldots, g^*_\rho)$ and $\Omega(x_1, \ldots, x_\rho)$, is itself an algebraic extension of $\Omega(g^*_1, \ldots, g^*_\rho)$, while $\Omega(x_1, \ldots, x_\rho)$ is an

[15] The transfer principle is formulated correspondingly for every basis question treated here.

[16] E. Steinitz, Algebraische Theorie der Körper, § 24, Crelle's Journal 137 (1909).

[17] Compare, for example, Weber, Lehrbuch d. Algebra 1, § 144 (1st ed.) or § 55 (small edition).

algebraic extension of $\mathfrak{K}^*_{\rho\rho}$. Thus $\mathfrak{K}^*_{\rho\rho}$ arises from $\Omega(g_1^*, \ldots, g_\rho^*)$ by adjoining a finite number of elements from $\mathfrak{K}^*_{\rho\rho}$, or also by adjoining one suitably chosen function; i.e. one has

$$\mathfrak{K}^*_{\rho\rho} = \Omega(g_1^*, \ldots, g_\rho^*, g_{\rho+1}^*, \ldots, g_k^*) = \Omega(g_1^*, \ldots, g_\rho^*, g_0^*).$$

By the transfer principle we therefore have

Theorem II: Every field $\mathfrak{K}_{n\rho}$ possesses a rational basis of algebraic rank ρ; the rational basis can in particular be chosen so that it contains at most $\rho + 1$ functions.

Let an arbitrary system of ρ algebraically independent functions from $\mathfrak{K}_{n\rho}$ be given by $h_1(x), \ldots, h_\rho(x)$, and suppose it can be extended by the above method to a rational basis

$$h_1(x), \ldots, h_\rho(x); h_{\rho+1}(x), \ldots, h_\sigma(x).$$

By the defining property, relations exist:

$$\begin{aligned} h_i(x) &= \chi_i(g_1(x), \ldots, g_k(x)) \quad (i = 1, 2, \ldots, \sigma), \\ g_j(x) &= \psi_j(h_1(x), \ldots, h_\sigma(x)) \quad (j = 1, 2, \ldots, k), \end{aligned}$$

where χ_i and ψ_j denote rational functions of their arguments. Thus:

From any rational basis one obtains every further rational basis by a birational transformation; and:

If

$$h_1(x), \ldots, h_\rho(x)$$

is any system of ρ algebraically independent functions from $\mathfrak{K}_{n\rho}$, then $\mathfrak{K}_{n\rho}$ is a finite algebraic extension of

$$\Omega(h_1(x), \ldots, h_\rho(x)).$$

§ 5. Involution form and involution basis of the fields $\mathfrak{K}_{n\rho}$.

Whereas every rational basis constructed in § 4 had the disadvantage of depending on the arbitrarily chosen starting functions, we now show the existence of a *rational basis uniquely determined by the field.*

By § 4, $\Omega(x_1, \ldots, x_\rho)$ is an algebraic extension — say of degree i — of $\mathfrak{K}^*_{\rho\rho}$, and, because

$$\mathfrak{K}^*_{\rho\rho}(x_1, \ldots, x_\rho) = \Omega(x_1, \ldots, x_\rho),$$

arises by adjoining the elements $x_1, \ldots, x_\rho$, which are algebraic with respect to $\mathfrak{K}^*_{\rho\rho}$. We therefore consider the integral function $\Phi^*(z, u)$ irreducible with respect to $\mathfrak{K}^*_{\rho\rho}$, with the zero

$$z = u_1 x_1 + u_2 x_2 + \cdots + u_\rho x_\rho = u(x),$$

which is of ith degree in z. Further, as the product of the quantities conjugate to $[z - u(x)]$, $\Phi^*(z, u)$ is a homogeneous form of dimension i in $z, u_1, \ldots, u_\rho$; hence as coefficient of z^i it contains a function from $\mathfrak{K}^*_{\rho\rho}$ that is free of u. If we make this highest coefficient equal to 1 by division, then the irreducible function $\Phi^*(z, u)$ is uniquely determined. The homogeneous form so defined,

$$\begin{aligned} \Phi^*(z, u) = z^i + {\sum}' g^*_{\alpha\alpha_1\ldots\alpha_\rho}(x_1, \ldots, x_\rho) z^\alpha u_1^{\alpha_1} \cdots u_\rho^{\alpha_\rho}, \\ (\alpha + \alpha_1 + \cdots + \alpha_\rho = i), \end{aligned} \tag{1}$$

is to be called the *involution form*[18] of $\mathfrak{K}^*_{\rho\rho}$; from (1) one obtains

$$\begin{aligned} \Phi(z, u) = z^i + {\sum}' g_{\alpha\alpha_1\ldots\alpha_\rho}(x) z^\alpha u_1^{\alpha_1} \cdots u_\rho^{\alpha_\rho}, \\ (\alpha + \alpha_1 + \cdots + \alpha_\rho = i), \end{aligned} \tag{2}$$

as an *involution form* of $\mathfrak{K}_{n\rho}$.

The coefficients of the involution form $\Phi^*(z, u)$ will turn out to be a rational basis of $\mathfrak{K}^*_{\rho\rho}$; i.e. the relation

$$\Omega(g^*_{\alpha\alpha_1\ldots\alpha_\rho}(x_1, \ldots, x_\rho)) = \mathfrak{K}^*_{\rho\rho} \quad (\alpha + \alpha_1 + \cdots + \alpha_\rho = i) \tag{3}$$

[18] The reason for the name will be clear at the end of the paragraph; $\sum'$ means that the summation is to extend over all terms except z^i.

holds. For the proof we observe that $\Phi^*(z,u)$ has coefficients from $\Omega(g^*_{\alpha\alpha_1\ldots\alpha_\rho}(x))$ and, because of its irreducibility with respect to $\mathfrak{K}^*_{\rho\rho}$, is all the more irreducible with respect to this subfield. Hence $\Phi^*(z,u)$ gives the irreducible equation of $u(x)$ with respect to $\Omega(g^*_{\alpha\alpha_1\ldots\alpha_\rho}(x))$. It follows that $\Omega(x_1,\ldots,x_\rho)$ is an algebraic extension — of degree i — of $\Omega(g^*_{\alpha\alpha_1\ldots\alpha_\rho}(x))$. Since $\mathfrak{K}^*_{\rho\rho}$ lies between $\Omega(g^*_{\alpha\alpha_1\ldots\alpha_\rho}(x))$ and $\Omega(x_1,\ldots,x_\rho)$, the identity of degrees is known to imply the identity of the fields, hence relation (3). By the transfer principle, the coefficients of $\Phi(z,u)$ correspondingly yield a rational basis of $\mathfrak{K}_{n\rho}$; i.e. we have

Theorem III: The coefficients of the involution form constitute a rational basis determined by the field, the involution basis; for fields $\mathfrak{K}^*_{\rho\rho}$ this basis is uniquely determined.[19]

We add an irrational interpretation of this fact, which establishes the connection with the geometric theory of involutions.

The factorization of $\Phi^*(z,u)$ into linear factors is given in an extension field by

$$\begin{aligned}\Phi^*(z,u) = (z-u_1x_1-\cdots-u_\rho x_\rho)(z-u_1x_1^{(1)}-\cdots-u_\rho x_\rho^{(1)})\cdots\\ \cdot(z-u_1x_1^{(i-1)}-\cdots-u_\rho x_\rho^{(i-1)}).\end{aligned} \tag{4}$$

Comparison with (1) shows that the functions

$$g^*_{\alpha\alpha_1\ldots\alpha_\rho}(x_1,\ldots,x_\rho) \quad (\alpha+\alpha_1+\cdots+\alpha_\rho=i)$$

are identical with the elementary symmetric functions of the rows of quantities

$$\begin{matrix} x_1 & x_2 & \cdots & x_\rho \\ x_1^{(1)} & x_2^{(1)} & \cdots & x_\rho^{(1)} \\ \vdots & \vdots & & \vdots \\ x_1^{(i-1)} & x_2^{(i-1)} & \cdots & x_\rho^{(i-1)}. \end{matrix} \tag{5}$$

Thus every function from $\mathfrak{K}^*_{\rho\rho}$ is a rational combination of these elementary functions, symmetric in the rows (5), and conversely every symmetric function of the rows (5) belongs to $\mathfrak{K}^*_{\rho\rho}$.

Thus, as

$$f^*(x) = \frac{F^*(x)}{H^*(x)}$$

runs through all functions from $\mathfrak{K}^*_{\rho\rho}$, there is a system of infinitely many relations:

$$\frac{F^*(x)}{H^*(x)} = \frac{F^*(x^{(1)})}{H^*(x^{(1)})} = \cdots = \frac{F^*(x^{(i-1)})}{H^*(x^{(i-1)})}, \tag{6}$$

or, summarized,

$$F^*(x^{(k)})H^*(x^{(l)}) - F^*(x^{(l)})H^*(x^{(k)}) = 0 \quad (k,l=0,1,\ldots,i-1),$$

where neither $F^*(x^{(k)})$ nor $H^*(x^{(k)})$ vanishes identically, since they are conjugate to $F^*(x)$ and $H^*(x)$ and formally symmetric.

Conversely, for every row of quantities ξ satisfying the infinitely many relations

$$F^*(x)H^*(\xi) - H^*(x)F^*(\xi) = 0, \qquad H^*(\xi) \neq 0, \tag{7}$$

one obtains membership in the rows of quantities (5).

Indeed, if $G^*(x)$ denotes the common denominator of the coefficients of $\Phi^*(z,u)$, then by assumption (7) the function

$$G^*(\xi)\cdot\Phi^*(z,u;\xi),$$

which is also integral in ξ, does not vanish identically — that is, the coefficients of $\Phi^*(z,u;\xi)$ do not become indeterminate — and has the zero

$$z = u_1\xi_1 + u_2\xi_2 + \cdots + u_\rho\xi_\rho.$$

Because of (7), $\Phi^*(z,u;x)$ therefore also has the zero $z=u(\xi)$; i.e. ξ belongs to the rows of quantities (5). The corresponding assertion holds by (6) if ξ satisfies a system of equations with respect to a conjugate row of quantities:

$$F^*(x^{(k)})H^*(\xi) - H^*(x^{(k)})F^*(\xi) = 0; \qquad H^*(\xi)\neq 0.$$

[19] For fields $\mathfrak{K}_{n\rho}$ the involution form may depend on the mapping; uniqueness can be obtained by a mapping with indeterminate coefficients.

Thus the rows of quantities (5) are defined as the solutions ξ of the system of equations

$$F^*(x)H^*(\xi) - H^*(x)F^*(\xi) = 0; \qquad H^*(\xi) \neq 0,$$

where $F^*(x) : H^*(x)$ runs through all functions from $\mathfrak{K}^*_{\rho\rho}$; here the row x may also be replaced by any conjugate row.

Geometrically this says the following, interpreting each row x as a point: the solutions of (7) represent, in a ρ-dimensional linear space, ∞^ρ groups of i points each, in such a way that any point of the group determines the individual group uniquely. Such a system of point-groups is called an "*involution in a linear space*."[20]

Correspondingly, a field $\mathfrak{K}_{n\rho}$ characterizes an involution in a linear space consisting of curves, surfaces, and so on.

By the preceding, the degree i of $\Omega(x_1, \ldots, x_\rho)$ with respect to $\mathfrak{K}^*_{\rho\rho}$ is equal to the number of solution systems of (7) satisfying the side condition $H^*(\xi) \neq 0$ — which we may call the solution systems of $\mathfrak{K}^*_{\rho\rho}$. Hence the argument used to prove the existence of the involution basis also permits the following irrational formulation:

"If $\mathfrak{L}^*$ is a divisor of $\mathfrak{K}^*_{\rho\rho}$ and every solution system of the field $\mathfrak{L}^*$ is also a solution system of $\mathfrak{K}^*_{\rho\rho}$, then $\mathfrak{L}^*$ and $\mathfrak{K}^*_{\rho\rho}$ are identical."

The corresponding statement holds, by the transfer principle, for divisors $\mathfrak{L}$ of $\mathfrak{K}_{n\rho}$.

§ 6. Minimal basis of the fields $\mathfrak{K}_{n\rho}$.

This paragraph gives an overview of known theorems, but in an expanded formulation made possible by the transfer principle and the existence of the rational basis.

Definition IV: A rational basis of $\mathfrak{K}_{n\rho}$ is called a minimal basis if it contains only the minimum number ρ of functions, which must then be algebraically independent.

From theorems of Lüroth, Castelnuovo, and Enriques[21] one obtains the facts:

I. Every field $\mathfrak{K}_{n1}$ possesses a minimal basis, the Lüroth function of the field, uniquely determined up to linear fractional transformation.

II. For fields $\mathfrak{K}_{n2}$, a minimal basis exists always, and in general only, if one permits an algebraic extension of the coefficient field Ω.

III. For fields $\mathfrak{K}_{n\rho}$ ($\rho \geq 3$), in general no minimal basis exists. A characterization of the special fields $\mathfrak{K}_{n\rho}$ with minimal basis has not yet been attempted.

We briefly indicate the proof of Lüroth's theorem for fields $\mathfrak{K}^*_{11}$ given by E. Steinitz:[22]

Let $\Phi^*(z, u)$ be the involution form of $\mathfrak{K}^*_{11}$, and let $\psi^*(x)$ be any coefficient of $\Phi^*(z, u)$ that actually contains the indeterminate x. It turns out that the degree of $\Omega(x)$ with respect to $\Omega(\psi^*(x))$ is equal to the degree of $\Omega(x)$ with respect to $\mathfrak{K}^*_{11}$, whence the identity of the fields follows. Further, if $\Omega(\chi^*(x))$ is equal to $\Omega(\psi^*(x))$, then the mutual rational dependence of $\chi^*(x)$ and $\psi^*(x)$ implies the linear fractional dependence of the two functions.

By the transfer principle, Lüroth's theorem therefore also admits the formulation:

The minimal basis of the fields $\mathfrak{K}_{n1}$ consists of any function of the involution basis; and any two functions of the involution basis of $\mathfrak{K}_{n1}$ depend on one another by a linear fractional transformation.

G. Castelnuovo proves Fact II, by means of the geometric theory of involutions, for fields $\mathfrak{K}^*_{22}$ derived from a rational basis of three functions. Since the transfer principle also remains valid under finite algebraic extension of the coefficient field Ω, the existence of the rational basis makes the proof complete for all fields $\mathfrak{K}_{n2}$.

Regarding III, it should be noted that Enriques constructs a nonrational involution in three-dimensional space; that is, a field $\mathfrak{K}_{33}$ which has no minimal basis.

[20]The excluded solutions of (7), for which $F^*(\xi) = 0$, $H^*(\xi) = 0$, and likewise the excluded solutions of the system of equations corresponding to the involution basis, are called fundamental points of the involution; this also includes those arising by homogenization, the points "lying at infinity."

[21]J. Lüroth: Beweis eines Satzes über rationale Kurven, Math. Ann. 9 (1875). — G. Castelnuovo: Sulla razionalità delle involuzioni piane, Math. Ann. 44 (1893). — F. Enriques: Sopra una involuzione non razionale dello spazio, Rend. Acc. Linc., Vol. 21, 21 Jan. 1912.

[22]Loc. cit., § 24.